ZKP IN

POST- QUANTUM ERA

HACKATHON 2026

STATISTICAL INDIAN INSTITUTE
UNITY IN DIVERSITY

FORTYTWO
SECURE TODAY. FUTURE READY.

Agenda

- What is Post Quantum Cryptography ?
- Post Quantum Cryptographic Algorithms
- Post Quantum Migration in Industry
- What is ZKP?
- Requirements and Components of ZKP
- Vulnerable Protocols in ZKP
- Vulnerable ZKP Protocols Deployed in Industry
- The HNDL Risk
- Applications of ZKP at HIGH Risk
- Quantum Safe alternatives in ZKP
- Current Research on PQ-ZKP
- Quantum migration of ZKP
- Key Takeaways
- Links for interested readers

Copyright © Fortytwo Labs

What is Post Quantum Cryptography ?

HARVEST NOW, DECRYPT LATER

- **Digital security** is built on the 5 pillars – Confidentiality, Integrity, Availability, Authentication, Non-repudiation
- **Encryption** and **Digital signature** protocols ensure a secure network based on these pillars
- Security in the digital world is based on the mathematical hardness of the classical cryptographic algorithms – **RSA** and **ECC**

Copyright © Fortytwo Labs

What is Post Quantum Cryptography ?

Threat for RSA

- RSA is an asymmetric encryption algorithm
- The public key is a 2048-bit integer and the private key is its prime factor
- The computational complexity for prime factorization of any n-bit integer is,

$$e^{(c+o(1))(\ln n)^{1/3}(\ln \ln n)^{2/3}}$$

- In 1994, Peter Shor presented an algorithm which can determine prime factors of an integer in real time on a powerful machine

Threat for ECC

- Elliptic curve cryptography is based on elliptic curve theory and Diffie-Hellman key exchange
- The public key sizes are much smaller than RSA
- The computational complexity lies in solving the discrete logarithm problem (DLP) in a large finite cyclic group, G : given $g, p(large\ prime), g^a mod\ p$, it is difficult to find a.
- Shor's algorithm can make $\Omega(\log|G|)$ group operation queries in polynomial time

Copyright © Fortytwo Labs

What is Post Quantum Cryptography ?

Classical Cryptography vs Quantum Cryptography

- Classical cryptography uses mathematically hard problems to derive security protocols to protect information
- The mathematical hardness is defined with respect to the computational power of classical computers, which are based on 0/1 bit
- The encryption or signature is generated using public/private key pairs which cannot be cracked in real time due to the low computing power of a classical computer
- A quantum computer is built on qubits which are dependent on quantum states
- This increases the computational power by many folds, hence algorithms like Shor's can be implemented in real time
- This creates a risk to the existing classical cryptography

Copyright © Fortytwo Labs

What is Post Quantum Cryptography ?

Quantum Cryptography Timeline

2001 - researchers at IBM implemented Shor's algorithm on a 7-qubit NMR computer

2011 – D-Wave systems claimed to produce first commercial quantum computer

2016 – IBM launched first cloud based quantum computing platform IBM-Q

2016 – NIST initiated a standardization process to find quantum-safe algorithms for key exchange and signature

2020 to 2023 – Major advancements in building quantum machines with higher qubit count and error correction by IBM, Google, Microsoft

2025 – NIST standardized 3 protocols for key exchange and signature

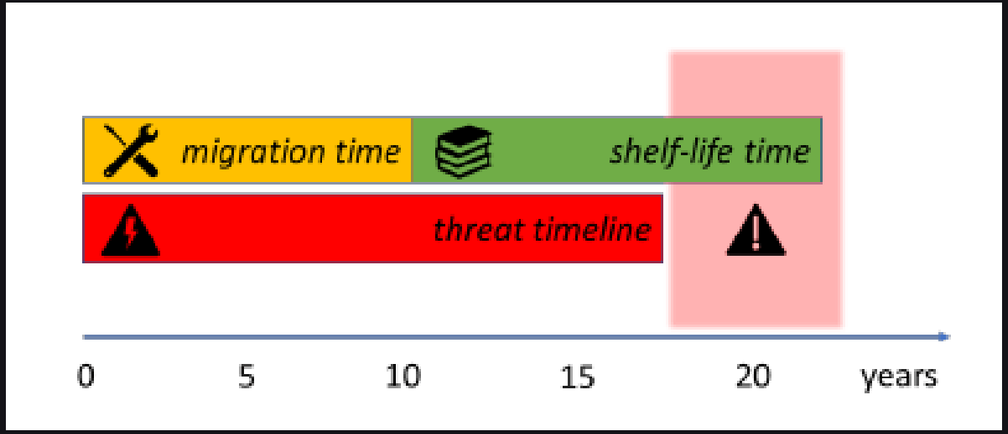
Q-Day – Experts say, a noise free quantum computer which can break a 2048-bit encryption, may be available as early as **2030**

Copyright © Fortytwo Labs

What is Post Quantum Cryptography ?

Mosca's Theorem

*If **X** represents the number of years that data must be kept secure, and Y is the estimated time needed to complete the transition, then organizations must start transitioning to post-quantum algorithms before **X + Y** exceeds the expected time **Z** for a cryptographically relevant quantum computer to be built.*



Key Establishment Scheme	Parameters	Transition
Finite Field DH and MQV [SP80056A]	112 bits of security strength	***Deprecated*** after 2030 ***Disallowed*** after 2035
	≥ 128 bits of security strength	***Disallowed*** after 2035
Elliptic Curve DH and MQC [SP80056A]	112 bits of security strength	***Deprecated*** after 2030 ***Disallowed*** after 2035
	≥ 128 bits of security strength	***Disallowed*** after 2035
RSA [SP80056B]	112 bits of security strength	***Deprecated*** after 2030 ***Disallowed*** after 2035
	≥ 128 bits of security strength	***Disallowed*** after 2035

Digital Signature Algorithm Family	Parameters	Transition
ECDSA [FIPS186]	112 bits of security strength	***Deprecated*** after 2030 ***Disallowed*** after 2035
	≥ 128 bits of security strength	***Disallowed*** after 2035
EdDSA [FIPS186]	≥ 128 bits of security strength	***Disallowed*** after 2035
RSA [FIPS186]	112 bits of security strength	***Deprecated*** after 2030 ***Disallowed*** after 2035
	≥ 128 bits of security strength	***Disallowed*** after 2035

Copyright © Fortytwo Labs

Post Quantum Cryptographic Algorithms

Alternative areas for post quantum cryptographic algorithms

- **Lattice Based** – hard geometric challenge to find the shortest vector in a high-dimensional lattice
- **Hash based** – rely purely on the security of cryptographic hash functions
- **Code based** – difficulty of decoding random linear error-correcting codes
- **Multivariate based** – solving complex systems of nonlinear equations
- **Isogeny based** – elliptic curve theory

Copyright © Fortytwo Labs

Post Quantum Cryptographic Algorithms

FIPS 203

Federal Information Processing Standards Publication

Module-Lattice-Based Key-Encapsulation Mechanism Standard

Category: Computer Security Subcategory: Cryptography

Information Technology Laboratory
National Institute of Standards and Technology
Gaithersburg, MD 20899-8900

This publication is available free of charge from:
https://doi.org/10.6028/NIST.FIPS.203

Published August 13, 2024

U.S. Department of Commerce
Gina M. Raimondo, Secretary

National Institute of Standards and Technology
Laurie E. Locascio, NIST Director and Under Secretary of Commerce for Standards and Technology

FIPS 204

Federal Information Processing Standards Publication

Module-Lattice-Based Digital Signature Standard

Category: Computer Security Subcategory: Cryptography

Information Technology Laboratory
National Institute of Standards and Technology
Gaithersburg, MD 20899-8900

This publication is available free of charge from:
https://doi.org/10.6028/NIST.FIPS.204

Published August 13, 2024

U.S. Department of Commerce
Gina M. Raimondo, Secretary

National Institute of Standards and Technology
Laurie E. Locascio, NIST Director and Under Secretary of Commerce for Standards and Technology

FIPS 205

Federal Information Processing Standards Publication

Stateless Hash-Based Digital Signature Standard

Category: Computer Security Subcategory: Cryptography

Information Technology Laboratory
National Institute of Standards and Technology
Gaithersburg, MD 20899-8900

This publication is available free of charge from:
https://doi.org/10.6028/NIST.FIPS.205

Published: August 13, 2024

U.S. Department of Commerce
Gina M. Raimondo, Secretary

National Institute of Standards and Technology
Laurie E. Locascio, NIST Director and Under Secretary of Commerce for Standards and Technology

NIST standards, 2025

Copyright © Fortytwo Labs

Post Quantum Cryptographic Algorithms

Threat for Hash functions

- Hash functions, $\mathcal{H}: X \rightarrow \{0,1\}^n$ map a message into a fixed length output so that, given the, $\mathcal{H}(x)$ it is hard to determine the pre-image in real time

- Searching the pre-image by brute force is of $O(n)$

- In 1996, Lov Grover came up with an algorithm which reduces the search to $O(\sqrt{n})$

- Advent of quantum computers would give rise to brute-force attacks and collision attacks due to this quadratic speed-up

Copyright © Fortytwo Labs

Post Quantum Cryptographic Algorithms

Hash/XOF Algorithm Family	Variants	Collision Security Strength	Collision Security Category	Preimage Security Strength	Preimage Security Category
SHA-1 [FIPS180]	SHA-1	80 bits	< 1	160 bits	1
SHA-2 [FIPS180]	SHA-224 SHA-512/224	112 bits	< 1	224 bits	3
	SHA-256 SHA-512/256	128 bits	2	256 bits	5
	SHA-384	192 bits	4	384 bits	5
	SHA-512	256 bits	5	512 bits	5
SHA-3 [FIPS202]	SHA3-224	112 bits	< 1	224 bits	3
	SHA3-256	128 bits	2	256 bits	5
	SHAKE128	128 bits	2	128 bits	2
	SHA3-384	192 bits	4	384 bits	5
	SHA3-512	256 bits	5	512 bits	5
	SHAKE256	256 bits	5	512 bits	5

Copyright © Fortytwo Labs

Post Quantum Migration in Industry

MIGRATE FROM RSA, ECC BASED CRYPTOGRAPHY

TO QUANTUM SAFE CRYPTOGRAPHY

By 2028

Organizations are expected to define their migration plans

By 2030

High-priority services and critical national infrastructures should complete PQC migration

By 2035

Target full migration across all systems, services and products globally

Copyright © Fortytwo Labs

What is ZKP?

- **Zero – knowledge proofs (ZKPs)** are cryptographic proofs that allow to prove knowledge of something without revealing the thing itself

- There are 2 entities involved – a **prover** (one who proves the knowledge of secret) and a **verifier** (one who validates the proof without learning the secret)

- **Interactive ZKP** – requires multiple rounds of communication between the prover and verifier where the verifier sends challenges to the prover and receives his responses

- **Non-Interactive ZKP** – a single message from prover to verifier; the verifier can check proof independently (used for trustless verification on decentralized system)

Copyright © Fortytwo Labs

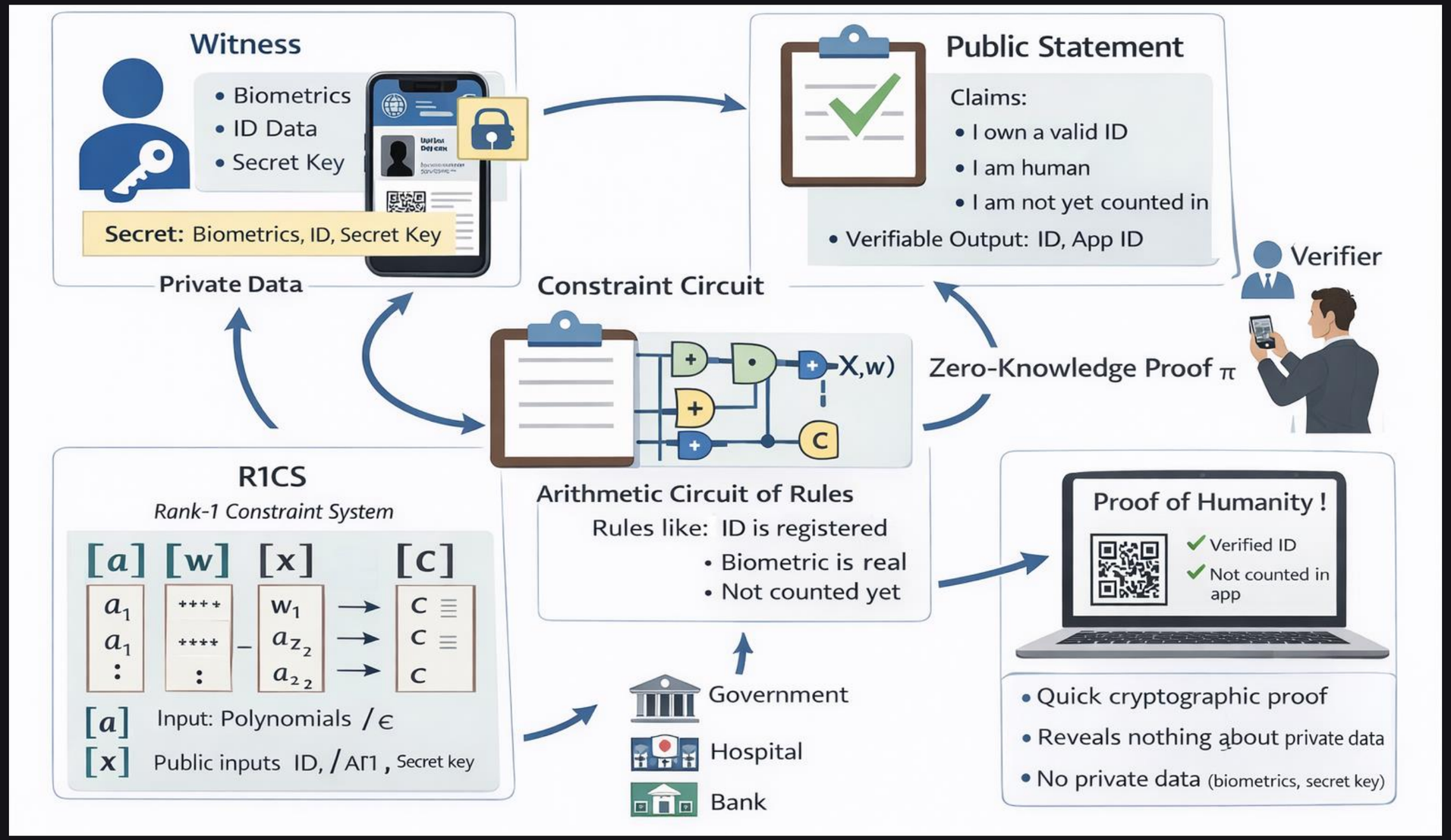
Requirements and Components of ZKP

- **Completeness**: if input is valid, then proof must always convince the verifier
- **Soundness**: it must be practically impossible for a prover to trick the verifier with invalid input
- **Zero-knowledge**: the verifier learns nothing about the secret from the proof

- **Witness**: the secret information that the prover knows and uses to demonstrate the proof
- **Claim**: the claim that the prover is making about the witness
- **Constraints**: mathematical rules or conditions that need to be true to make the proof valid
- **Circuits**: structured representation of the constraints; the prover uses the circuit to demonstrate they know a set of inputs that satisfy all constraints without revealing the inputs
- **Rank 1 Constraint System (R1CS)**: algebraic circuit expressed as a set of vectors and matrices and then converted to polynomials

Copyright © Fortytwo Labs

Requirements and Components of ZKP



Copyright © Fortytwo Labs

Vulnerable Protocols in ZKP

Sl. No	ZKP Protocols/Frameworks	Mathematical Primitive	Main feature of ZKP	Is it Quantum Secure?
1	Σ-protocol (like Schnorr)	Discrete log in cyclic group, G	Knowledge soundness	Broken by Shor
2	Fiat-Shamir NIZK	Cryptographic Hash function (ROM/QROM)	Non-interactivity	Grover reduces security
3	Pederson Commitment based ZK	Group exponentiation (DLP)	Perfect hiding, computational binding	Broken by Shor
4	Groth16 zk-SNARK	Bilinear pairing	Constant-size proof	Broken by Shor
5	PLONK	Polynomial commitments and pairing	Setup	Broken by Shor
6	Bulletproofs	Elliptic curve inner-product	Logarithmic proof size	Broken by Shor

Copyright © Fortytwo Labs

Vulnerable ZKP Protocols Deployed in Industry

- A **decentralized blockchain platform** powered by the cryptocurrency ETH
- Uses ZKP to solve **scalability, privacy and efficiency challenges** in blockchain applications
- Ethereum is used in various business sectors like – **Decentralized Finance, NFT marketplaces, Gaming, Micropayments, decentralized identity**
- Ethereum uses ZKP protocols extensively:
  - **Groth16 zk-SNARK** for succinct proof verification
  - **PLONK based SNARK** for smart contract scaling
  - **zk-SNARK** for low-cost transactions
  - **zk-STARK** for scalable rollups

Copyright © Fortytwo Labs

Vulnerable ZKP Protocols Deployed in Industry

- The first cryptocurrency to develop zero-knowledge encryption for payments
- Uses ZKP mainly for authentication, scalability
- ZCash uses ZKP protocols extensively:
  - **Groth16 zk-SNARK** for privacy during transfers
  - **PHGR13 SNARK** for private transactions
  - **PLONKish SNARK** for no-trusted setup
  - **RedDSA(Schnorr)** for transaction signing

Copyright © Fortytwo Labs

Vulnerable ZKP Protocols Deployed in Industry

- A fully decentralized privacy-preserving blockchain built on the Ethereum platform
- Uses novel client-site ZKP to keep balances, trades private
- Aztec uses ZKP protocols extensively:
  - **PLONK based zk-SNARK** for transaction validity proof
  - **HASH-based commitments** for efficient on-chain verification

Copyright © Fortytwo Labs

Vulnerable ZKP Protocols Deployed in Industry

- Developed **World ID** which is a privacy-preserving identity protocol to prove you are a unique human, without revealing who you are (**Proof of Personhood**)
- World ID solves the problem of deepfakes and bots
- Allows users to prove name, biometric data, government ID
- World ID uses ZKP protocols extensively:
  - **Sephamore (Groth16 zk-SNARK + Merkle tree)** for membership validity proof
  - **Groth16 zk-SNARK** for efficient verification
  - **Merkle Tree** for identity commitments

Copyright © Fortytwo Labs

The HNDL Risk

Harvest

ZK Proofs, CRS, verification keys, public commitments, signature backed credentials, Identity authentication transcripts

Store

Harvested data stored for long-term

Wait

Wait till advent of scalable quantum computers

Decrypt

Signature schemes can be broken, commitment bindings will fail, knowledge soundness collapse

What leaks out?

Allow valid proof forging, enable fake identity claims, break non-repudiation guarantees

Produce valid-looking historical proofs without actually possessing a secret

Copyright © Fortytwo Labs

Applications of ZKP at HIGH Risk

- **Long-lived identity credentials**
- **Government digital ID systems**
- **Blockchain-based compliance proofs**
- **Attribute-based authentication systems**
- **Systems relying on pairing-based SNARKs**

HARVEST NOW, DECRYPT LATER

collect proof transcripts now, enable proof forgery and long-term authenticity failure later

Copyright © Fortytwo Labs

Applications of ZKP at HIGH Risk

Board of Governors of the Federal Reserve System

The Federal Reserve, the central bank of the United States, provides the nation with a safe, flexible, and stable monetary and financial system.

About the Fed | News & Events | Monetary Policy | Supervision & Regulation

Home > Economic Research > Finance and Economics Discussion Series

Finance and Economics Discussion Series

September 2025

"Harvest Now Decrypt Later": Examining Post-Qua
and the Data Privacy Risks for Distributed Ledger M

Jillian Mas

This case study analyzes the impact of PQC to a public, permissionless distributed ledger network like Bitcoin and concludes that the HNDL threat is (1) current and active within Bitcoin; (2) without significant data privacy mitigations for transactions completed before PQC is substantially deployed within the network; and (3) activated on an asynchronous timeline because of the varying strengths of the traditional cryptographic tools used within Bitcoin's decentralized networks. With sufficient technical know-how, a bad actor could join the Bitcoin network as a node operator, harvest a copy of the network's database (here, the blockchain ledger), independently store their ledger copy without PQC protection, and in the fullness of time pull their ledger replica into a quantum computing environment to crack the traditional cryptography, attack the protections of private keys, and reveal the blockchain ledger's data in plain text. The following sections discuss this process, and potential mitigations, in greater detail.

However, even if the community collectively, willingly, and fully adopted each of these mitigations, one of the three key vulnerabilities would remain without a full solution: data privacy. First, transitioning to only PQC address types, for example, does not retroactively protect a Bitcoin user's previous transactions with less–quantum re
could transfer their bitcoin funds from a legacy wallet to a PQC sec

users whose data privacy becomes vulnerable to a quantum computer. Data thought to be protected by blockchain hashing and cryptographic techniques could be revealed in some amount of detail, including plain text, diminishing the data privacy of a distributed network's previous
s to view private keys and transaction data that users

data with only a hard fork and *future* crypto agility, with the difficulty possibly increasing alongside a transaction's relative age. A bad actor storing a current replica of the Bitcoin ledger may need only wait long enough to obtain access to a sufficiently powerful quantum computer to reveal the ledger's cryptographically obscured transaction data, making Mosca's Theorem the

Copyright © FortyTwo Labs

Quantum Safe alternatives in ZKP

Sl.No	ZKP Protocols/Frameworks	Mathematical Primitive	Main feature of ZKP	Is it Quantum Secure?
1	zk-STARK	Hash functions and polynomials	Transparent setup	Yes, if Hash is secure
2	MPC-in-the-Head ZK	Symmetric cryptography + secret sharing	Transparent zero-knowledge	Yes, if Hash is secure
3	Lattice based ZK	LWE, SIS over $\mathbb{Z}_q^n$	Post quantum soundness	Yes
4	Merkle Tree based ZK	Collision resistant hash trees	Efficient commitments	Yes, if Hash is secure

Copyright © Fortytwo Labs

Current Research on PQ-ZKP

IEEE Access

Received 18 January 2025, accepted 20 February 2025, date of publication 26 February 2025, date of current version 12 March 2025.
Digital Object Identifier 10.1109/ACCESS.2025.3545943

TOPICAL REVIEW

PP-PQB: Privacy-Preserving in Post-Quantum Blockchain-Based Systems: A Systematization of Knowledge

BORA BUGRA SEZER[1,2], SEDAT AKLEYLEK[2], AND URFAT NURIYEV[3,4]
[1]Graduate School of Natural and Applied Science, Ege University, 35040 Izmir, Türkiye
[2]Institute of Computer Science, University of Tartu, 50409 Tartu, Estonia
[3]School of Engineering, Karabakh University, Khankendi AZ2600, Azerbaijan
[4]Institute of Control Systems, Ministry of Science and Education of the Republic of Azerbaijan, Baku AZ1141, Azerbaijan
Corresponding author: Sedat Akleylek (sedat.akleylek@ut.ee)

The work of Bora Bugra Sezer was supported by the Scientific and Technological Research Council of Türkiye (TÜBİTAK) under Grant 1059B192301085. The work of Sedat Akleylek was supported by the Estonian Research Council under Grant PRG2531.

2025

Lattice-Based Zero-Knowledge Proofs and Applications: Shorter, Simpler, and More General*

Vadim Lyubashevsky[1], Ngoc Khanh Nguyen[1,2], and Maxime Plançon[1,2]
[1] IBM Research Europe, Zurich
[2] ETH Zurich, Zurich

2022

oter and Communications Security (December 2024)
https://doi.org/10.1145/3658644.3690330
ISBN: 9798400706363

The LaZer Library: Lattice-Based Zero Knowledge and Succinct Proofs for Quantum-Safe Privacy

Vadim Lyubashevsky
IBM Research Europe
Zurich, Switzerland
vad@zurich.ibm.com

Gregor Seiler
IBM Research Europe
Zurich, Switzerland
gseiler@posteo.net

Patrick Steuer
IBM Research Europe
Zurich, Switzerland
ick@zurich.ibm.com

2024

A non-PCP Approach to Succinct Quantum-Safe Zero-Knowledge

Jonathan Bootle[1,3], Vadim Lyubashevsky[1], Ngoc Khanh Nguyen[1,2], and Gregor Seiler[1,2]
[1] IBM Research – Zurich, Switzerland
nkn@zurich.ibm.com, [0000-0001-8240-6167]
vad@zurich.ibm.com
[2] ETH Zurich, Switzerland
gseiler@inf.ethz.ch
[3] UC Berkeley, USA
jonathan.bootle@berkeley.edu, [0000-0003-3582-3368]

2020

Quantum-Safe Identity Verification using Relativistic Zero-Knowledge Proof Systems

Yao Ma, Wen Yu Kon, Jefferson Chu, Kevin Han Yong Loh, Kaushik Chakraborty and Charles Lim
Global Technology Applied Research, JPMorganChase

2025

Copyright © Fortytwo Labs

Quantum migration of ZKP

Possible replacements

Classical Component	PQ alternative
Groth16	Lattice based
PLONK	Hash based MPC-in-the-Head
Pederson Commitment	SIS-based commitment
EC-DSA	ML-DSA
BLS Signatures	SLH-DSA

Challenges in Migration

- Huge computational overhead as key sizes are much larger compared to RSA/ECC algorithms
- Causes performance degradation
- Migration would be complex process
- Hardware acceleration requirements
- Increase in proof sizes and proving time
- Higher cost for verification

Copyright © Fortytwo Labs

Key Takeaways

- Quantum computers threaten classical public-key cryptography
- ZKP systems must migrate to PQ assumptions
- Lattices and hash-based constructions are promising
- Efficiency remains the main bottleneck
- Hybrid systems may dominate near-term deployments

Copyright © Fortytwo Labs

Links for interested readers

1. https://www.research-collection.ethz.ch/server/api/core/bitstreams/0ca4ae70-a1e9-419f-b574-07991439974c/content
2. https://www.youtube.com/watch?v=dxVVNuJZMiI
3. https://www.youtube.com/watch?v=xEDZ4tyesMY
4. https://eprint.iacr.org/2025/2099.pdf

Copyright © Fortytwo Labs

Towards future-ready ZKP…

Copyright © Fortytwo Labs